

# Security incident report

## Section 1: Identify the network protocol involved in the incident

The event involves sending requests to a DNS server and establishing TCP connections with two separate websites; therefore, the network protocols involved are TCP, HTTP, and DNS.

Primary protocol: HTTP. Supporting protocols: DNS and TCP.

## Section 2: Document the incident

This incident came to light after several users reported an anomaly on the website. When they visited the website, they were prompted to download and run a file that contained access to new recipes. Their personal computers have been operating slowly ever since.

Upon discovering the issue, the website owner attempted to log in to the administrative backend but was unable to do so, prompting them to contact the hosting provider. A sandbox environment was subsequently created to observe the suspicious website's behavior. It was observed that when a user logged in, the site established a normal connection with `yummyrecipesforme.com` before downloading a file and redirecting the user to `greatrecipesforme.com`.

Detailed network data from the sandbox session was captured using the network protocol analyzer `tcpdump`, and a thorough investigation of the website's scripts was conducted. The cybersecurity analyst inspected the tcpdump log and observed that the browser initially requested the IP address for the yummyrecipesforme.com website. Once the connection with the website was established over the HTTP protocol, the analyst recalled downloading and executing the file. The logs showed a sudden change in network traffic as the browser requested a new IP address for the greatrecipesforme.com URL. The network traffic was then rerouted to the new IP address for the greatrecipesforme.com website. The investigation ultimately confirmed that the web server had been compromised by a brute-force attack. Final confirmation of this happened when a former employee/hacker gained access to the web host via a brute-force attack and altered the administrator

password. They also embedded a JavaScript function in the source code designed to trick visitors into downloading and executing a file that redirected them to a malicious website.

### **Section 3: Recommend one remediation for brute force attacks**

To prevent such incidents from recurring, the following measures should be implemented:

Use MFA to verify user identities:

One of the authentication factors in MFA is unique, maximizing security.

Strengthen password policies, such as requiring strong passwords and mandatory periodic password changes:

These measures reduce the success rate of brute-force attacks and increase the cost of attacks for adversaries.